1. Clinical Question & Core Hypothesis

Clinical Question

What molecular mechanisms determine whether immune structures translate into clinical benefit?

Working Hypothesis

In bladder cancer, high intratumoral TIL density will predict favorable response to anti-PD-1/PD-L1 therapy only when TGF-β signaling is low and/or MSH6 is deficient; when TGF-β signaling is high, the benefit of TIL density is diminished or abolished, even with MSH6 deficiency. Therefore, the strongest responses to ICIs are expected in tumors that (a) exhibit high TIL density, (b) show low TGF-β pathway activity (e.g., low TGFB1 signaling and/or low pSMAD2/3 activity), and (c) harbor MSH6 loss or dysfunction leading to higher neoantigen burden. Clinically, this predicts higher objective response rates, longer disease control, and longer progression-free survival in the TIL-high, MSH6-deficient, low-TGF-β subgroup compared with TIL-high tumors with high TGF-β signaling or with intact MSH6. Secondary modifiers (CTCs, GREM1, urine DNA methylation markers, Immunoscore components) will be evaluated exploratively to refine risk stratification and to identify noninvasive readouts that mirror the core biology.

Element	Value
Imaging feature	Tumor-Infiltrating Lymphocyte (TIL) density
Anchor gene(s)	TGF-β, MSH6
Primary endpoint	therapy_response

2. Why This Gene — Not Others

Primary hypothesis gene(s)

TGF-β: TGF-β (core path 1) because it mechanistically enforces immune exclusion, ECM remodeling and T-cell dysfunction; (b) MSH6 (core path 2) because MMR deficiency increases neoantigen load and should enhance immunogenicity.

MSH6: MSH6 (core path 2) because MMR deficiency increases neoantigen load and should enhance immunogenicity.

Alternative / secondary genes considered

Tier	Gene	Role / Reason excluded from core
Secondary	CTCs	Supporting modifier from knowledge graph

Secondary	GREM1	Supporting modifier from knowledge graph
Secondary	urine_DNA_methylation_markers	Supporting modifier from knowledge graph
Secondary	CD8A	Supporting modifier from knowledge graph

3. Clinically Testable Predictions

Prediction 1: directly testable in bladder cancer cohorts and aligns with actionable clinical readouts. Experimental approach:

Prediction 2: Retrospective/prospective cohorts of ICI-treated bladder cancer with available tumor tissue.

Prediction 3: Measure core variables: TIL density (digital pathology and Immunoscore CD3/CD

4. Minimum Viable Validation Plan

A. Retrospective Cohort Analysis

Data source: TCGA-BLCA (Firehose Legacy), ~400 patients - Cox PH model with interaction term for the imaging feature and anchor gene - Kaplan-Meier 2x2 stratification

B. Tissue-Level Validation

Methods: Multiplex IF / IHC / spatial transcriptomics

Key readouts: - Immune structure markers (e.g., CD20, CD3, PNAd) - Effector cell localization (CD8) relative to tumor nests - Anchor gene protein / phospho-protein expression - Stromal features (collagen density by SHG, if applicable)

Goal: Confirm that the proposed mechanism operates at the spatial / tissue level.

C. Functional Validation

Models: Ex vivo organotypic cultures, 3D co-culture, syngeneic mouse models

Perturbations: Genetic (KO/rescue) or pharmacologic modulation of anchor gene

Readouts: T-cell infiltration, cytotoxicity (granzyme B / IFN-gamma), tumor growth / survival

Goal: Establish causal relationship between anchor gene activity and immune efficacy.

5. Strengths, Caveats & Alternative Directions

Strengths

- Interaction-first biomarker strategy — asks WHY the imaging feature sometimes fails, not just whether it predicts outcome.

Caveats & Risks

- Gene expression does not equal protein activity; phospho-protein or functional assays are needed for confirmation.

Alternative Directions

- CTCs: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.
- GREM1: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.